

Measuring Rhetorical Alarmism in Disaster News: A Quantitative Framework Applied to the 2024 Valencia Floods

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Abstract

1 Introduction

The floods that affected the Valencia region in October 2024 have been described as one of the most severe natural disasters in recent Spanish history, resulting in 228 fatalities and affecting thousands of people. In such contexts of uncertainty and disruption, news media play a crucial role in society, as they are responsible for disseminating information and shaping how events are publicly interpreted (Pantti et al. 2012; Cottle 2014). For this reason, media coverage contributes to the construction of meaning around disasters, influencing not only how they are understood but also how they are socially and politically responded to.

Disasters are not only objective events defined by their material consequences, but also socially constructed phenomena influenced by the media representations (Pantti et al. 2012). The way in which different media outlets select, frame and narrate these events, plays a critical role in defining their salience, perceived severity, and broader social implications. Rather than acting as neutral transmitters of information, media operate as *'amplification stations'* of risk, intensifying certain aspects of the event while downplaying others, thereby influencing public perception and collective responses (Kasperson et al. 1988). This can be further understood through the framing theory, which posits that media do not simply report events but actively construct narratives by selecting, emphasizing and organizing specific aspects of reality (Scheufele 1999). Frames operate by defining problems, attributing causes and suggesting moral evaluations or potential solutions (Entman 1993).

However, the process of risk amplification is not neutral and may lead to distortions in how crises are represented and understood. In highly mediatized disaster contexts, the selection and intensification of specific elements can give rise to narratives that emphasize threat, urgency, and uncertainty in disproportionate ways (Cottle 2012). Holm in (Holm 2012) listed an excellent example of this phenomenon regarding the media coverage of Katrina Hurricane, the *New York Times* wrote: *"America is once more plunged into a snakepit of anarchy, death, looting, raping, marauding thugs, suffering infrastructure, a gutted police force, insufficient troop levels, and criminally negligent government planning"*. This is particularly evident in alarmist headlines, where linguistic and rhetorical choices tend to condense and dramatize complex events, often privileging emotionally charged interpretations over contextual nuance. As noted by (Molek-Kozakowska 2013), headlines work as strategic devices that package information, activate emotional and cognitive responses, and frame the audience's understanding of the event. As a result, media coverage may contribute to the construction of alarmist

narratives that frame disasters as unprecedented, uncontrollable, or escalating phenomena, potentially amplifying public fear and shaping collective responses beyond the immediate material impact of the event (Altheide and Michalowski 1999; Altheide 2018).

Despite extensive research on media framing and risk communication in the context of natural disasters, existing studies tend to approach *alarmism* in broad or implicit terms, without providing a systematic framework to identify and quantify it (see next section). In particular, while previous work has examined emotional framing, fear appeals, sensationalism, and media hype in crisis contexts, relatively limited attention has been paid to the specific rhetorical patterns through which alarmist narratives are constructed. Importantly, not all negatively framed or emotionally charged headlines can be considered alarmist, highlighting the need for a more precise conceptualization of this phenomenon.

This article addresses this gap by introducing the concept of *rhetorical alarmism* as a multidimensional form of media framing that operates through identifiable linguistic and narrative structures. Building on existing literature on framing and risk amplification, the study proposes a framework to systematically capture how media discourse constructs disasters as exceptional, uncontrollable, or escalating events.

Focusing on the case of the 2024 Valencia floods (DANA), the article aims to identify and quantify the presence of *rhetorical alarmism* in news headlines across different Spanish media outlets and social media platforms using an automated algorithm and an ad-hoc index, identifying statistical differences between media types, and examine the temporal evolution of the index.

2 Literature Review

2.1 Media Coverage in Natural Disasters

Media play a central role in shaping public understanding of disasters, as well as in structuring broader aspects of social life. As argued by (Thompson 1995), communication media transform the spatial and temporal organization of social interaction, reconfiguring relations of visibility and power. In the context of disasters, this transformative role is particularly salient, as media do not simply transmit information about disruptive events but actively participate in their social construction.

In contemporary global communication environments, disasters can no longer be understood as isolated or purely natural phenomena. Instead, they are embedded within transnational communication flows that amplify their visibility and connect distant audiences. In this sense, disasters are increasingly *mediatized*, as media practices shape both the perception and the meaning of such events. Media effectively “enter into” disasters, shaping them from the *inside out* and the *outside in* by reconfiguring social relations and infusing events with cultural and political significance (Pantti et al. 2012; Cottle 2012).

This process is closely linked to the nature of news as discourse. Following (Van Dijk 1988), news should be understood not as a neutral representation of reality but as a form of public discourse that structures, interprets, and hierarchizes information. Through its linguistic, semantic, and rhetorical organization, news discourse selectively constructs meanings, embeds ideological perspectives, and guides the interpretation of events. In this sense, news does not merely reflect disasters but actively contributes to the construction of a shared *social imaginary*.

Within this framework, framing emerges as a key mechanism through which this construction operates. Framing refers to the process of selecting certain aspects of a perceived reality and making them more salient in communicative texts, in such a way that they promote a particular definition of the problem, a causal interpretation, a moral evaluation, and/or a treatment recommendation (Entman 1993). In the context of disasters, framing has significant implications for audience perception. As noted by (Chong and Druckman 2007), the frames adopted in media coverage can lead individuals to develop distinct interpretations of the same issue, influencing the relative weight assigned to different considerations in the formation of opinions.

Consequently, media coverage of disasters may vary substantially depending on how events are framed, shaping not only what is communicated but also how it is interpreted. Framing thus repre-

Media Type	Media Outlet
National Newspapers	<i>20 minutos, ABC, AS, EFE, El Mundo, El País EuropaPress, La Razón, La Vanguardia, Marca</i>
Local Newspapers	<i>Las Provincias, Levante EVM, El Mediterráneo, Torrent al día, Valencia Plaza</i>
National Radio	<i>La Cope, La Ser, Onda Cero</i>
Local TV	<i>7TeleValencia, À Punt, LaOchoTv</i>
National TV	<i>Antena 3, Telecinco, La Sexta, RTVE, Cuatro</i>
Social Media	<i>X, Facebook, Tiktok</i>

Table 1: Media outlets and social media included in the analysis

sents a key mechanism through which media discourse contributes to the construction of collective understandings of disasters.

While framing plays a decisive role in shaping how disasters are interpreted, media coverage also incorporates a strong emotional dimension. As argued by (Altheide 1997), the “problem frame” promotes a discourse of fear, understood as the pervasive communication, symbolic awareness, and expectation that danger and risk are central features of the social environment. In this sense, media discourse not only organizes the interpretation of events but also contributes to the affective construction of public problems. Previous research has shown that public understandings of social issues often incorporate the definitions, scenarios, and symbolic vocabularies circulated by the media, which become part of broader interpretative frameworks through which audiences make sense of risk and crisis.

This is particularly relevant in disaster contexts, where emotionalization can affect how risk, urgency, and severity are interpreted by the public. Moreover, media do not simply reflect risk but can amplify its social perception. According to (Kasperson et al. 1988), this amplification may occur through at least three main mechanisms: first, an increased volume of coverage may heighten the salience of a risk event, leading audiences to perceive it as more important, severe, or urgent; second, the presence of expert disagreement or public controversy may intensify uncertainty and concern; and third, risk signals may be amplified when they are communicated in emotionally intense or symbolically charged terms, such as catastrophic metaphors, alarmist language, or highly dramatic imagery. Together, these processes help explain how disaster coverage can move beyond information and toward intensified constructions of threat.

2.2 A framework for Rhetoric Alarmism

3 Materials and Methods

3.1 Corpus Construction

This study analyzes a dataset of 1,347 headlines from different official news sources. The aim is to examine the presence of ‘alarmist’ framing in the coverage of the DANA event. The unit of analysis defined is the *journalistic headline* and in the case of social media, a sample for first *n-words* is taken from the post. The study period covers from 29th October to 30 November 2024, separating the data into two periods: the first from 29th October to 4th November (*Impact*), and the second from 5th November to 30th November (*Post*). This division allows for a comparative analysis of the framing used in the immediate aftermath of the DANA event and in the subsequent weeks.

To ensure an appropriate representation of Spanish media ecosystem, the corpus includes headlines from a variety of sources, which include national and local scopes. The data set includes headlines listed on the Table 1 3.1:

Data collection followed a reproducible procedure based on structured search queries. Whenever possible, the *Advanced Google Search Engine* was used, combining boolean operators and site domains.

When this method was insufficient and provided poor results, internal search engines of media sites were used specifically in social media. A limitation was encountered in the case of *Instagram*, as the platform currently does not provide a reliable date-filtering mechanism comparable to search engines or other social media platforms such as *X* (Twitter). This constraint makes it difficult to reproduce the same sampling procedure, as result we decided to exclude this social media from the final corpus construction.

The general search pattern was as follows:

(dana OR "gota fría" OR lluvias OR torrenciales) AND Valencia AND 2024
site:<domain>

Explicit criteria were defined to ensure consistency across the headlines collected.

Inclusion Criteria

- Explicit reference to the phenomenon under study: DANA or heavy rainfall in Valencia
- Publication within the defined temporal window
- Headlines must originate from verifiable sources (URL or platform identifier)
- For social media: a minimum relevance threshold ($\geq 1,000$ interactions, such as likes or shares)

Exclusion Criteria

- Irrelevant or ambiguous content not directly related to the event
- Duplicated entries or republished content
- News outside the temporal scope

The sampling strategy consisted of selecting up to 50 relevant headlines per media outlet, ordered according to their position in search results and filtered based on the inclusion and exclusion criteria defined previously. This approach was adopted to ensure a comparable sample size across outlets, preventing larger media sources from being overrepresented relative to smaller or local outlets, and thereby enhancing the consistency and reproducibility of the analysis. In cases where a given outlet did not reach the target sample size, all available headlines meeting the selection criteria were retained.

A limitation of this approach is that the first set of sampled headlines does not necessarily correspond to equivalent temporal windows across media outlets. Due to differences in publication intensity over time, some outlets reached the target sample size more quickly than others. As a result, the collected samples may reflect not only outlet-specific framing strategies but also the natural temporal evolution of rhetorical alarmism.

Consequently, part of the observed differences in headline content may be driven by changes in the media discourse over time, rather than solely by differences between outlets. To address this issue, temporal variables such as time (in days) and coverage phase were incorporated into the statistical analysis. This allows for a more robust distinction between variations attributable to media-specific framing and those arising from the evolving phases of disaster reporting.

Following the data collection process, a structured database was constructed in which each observation corresponds to an individual headline. The final corpus comprises 1,347 headlines. For each observation, a set of variables was defined: the *media outlet* (source name), publication date, and the full headline text. Additionally, other categorical variables were incorporated, such as *media type* (e.g., press, radio, television, or social media) and the *temporal phase* of the event (impact or post-event stage). The dataset also includes an estimate of audience reach and the corresponding URL linking to the original content. The information related to audiences was collected from ...

3.2 Data Preprocessing

Text Normalization and Linguistic Processing

Prior to analysis, a text standardization process was applied to ensure consistency and reduce noise. The cleaning procedure included converting all text to lowercase, replacing punctuation with whitespace, collapsing multiple whitespace characters into a single space, and removing residual special characters. For social media content, additional preprocessing steps were implemented, including the removal of hashtags (#), mentions (@), emojis, and other non-linguistic symbols.

Once the text was normalized, all headlines were lemmatized to improve pattern detection. This process transforms each word (token) into its canonical base form, thereby minimizing the impact of inflectional variations such as verb conjugations, plural forms, and gender agreement. Lemmatization was performed using the UDPipe library (Straka and Straková 2017) with a Spanish pre-trained model (Taulé et al. 2008), which provides tokenization, part-of-speech tagging, and lemma extraction. Each headline was thus converted into a sequence of lemmas representing the original text. This lemmatized representation serves as the input for the subsequent regex-based detection process used to identify alarmist linguistic patterns.

Regex-based Algorithm for Detecting Rhetorical Alarmism

To classify news headlines according to the proposed framework, an automated algorithm was developed based on regular expression (*regex*) matching over lemmatized text. The construction of the regex patterns was guided by dictionaries associated with each of the four dimensions of *rhetorical alarmism* defined in this study. These dictionaries were manually elaborated and consist of keywords and linguistic expressions representative of each alarmism category. For each dimension, a set of regular expressions was generated to capture semantically and syntactically related patterns in the headlines. The algorithm evaluates each headline by matching its lemmatized tokens against the corresponding regex dictionaries. For every match, the associated alarmism category is recorded, the total number of detected alarmism categories is then computed for each headline.

To enable comparability across headlines of varying length, a normalized alarmism score is defined. Specifically, the number of detected alarmist patterns is divided by the total number of tokens in the headline, producing a dimensionless index of rhetorical alarmism (*RAI*). This normalization ensures that longer headlines are not systematically biased toward longer headlines, thereby allowing for fair comparisons across different media sources.

4 Conclusions

We conclude that....

External data

The experimental dataset is available at....

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References

- Altheide, David. 1997. "The News Media, The Problem Frame, and the Production of Fear." *The Sociological Quarterly* 38 (September): 647–668. <https://doi.org/10.1111/j.1533-8525.1997.tb00758.x>.
- . 2018. *Creating Fear: News and the Construction of Crisis*. May. <https://doi.org/10.4324/9780203794494>.
- Altheide, David and Michalowski, R. Sam. 1999. "FEAR IN THE NEWS." *The Sociological Quarterly* 40 (3): 475–503. <https://doi.org/10.1111/j.1533-8525.1999.tb01730.x>.
- Chong, Dennis and Druckman, James N. 2007. "A Theory of Framing and Opinion Formation in Competitive Elite Environments." *Journal of Communication* 57, no. 1 (March): 99–118. <https://doi.org/10.1111/j.1460-2466.2006.00331.x>.
- Cottle, Simon. 2012. "Mediatized Disasters in the Global Age: On the Ritualization of Catastrophe." January. <https://doi.org/10.1093/oxfordhb/9780195377767.013.10>.
- . 2014. "Rethinking media and disasters in a global age: What's changed and why it matters." *Media, War & Conflict* 7 (1): 3–22. <https://doi.org/10.1177/1750635213513229>.
- Entman, Robert M. 1993. "Framing: Toward clarification of a fractured paradigm." *Journal of Communication* 43 (4): 51–58.
- Holm, Isak Winkel. 2012. "The Cultural Analysis of Disaster." In *The Cultural Life of Catastrophes and Crises*, edited by Carsten Meiner and Kristin Veel, 15–32. Berlin, Boston: De Gruyter. <https://doi.org/doi:10.1515/9783110282955.15>.
- Kasperson, Roger E., Renn, Ortwin, Slovic, Paul, Brown, Halina S., Emel, Jacque, Goble, Robert, Kasperson, Jeanne X., and Ratick, Samuel. 1988. "The Social Amplification of Risk: A Conceptual Framework." *Risk Analysis* 8 (2): 177–187. <https://doi.org/10.1111/j.1539-6924.1988.tb01168.x>.
- Molek-Kozakowska, Katarzyna. 2013. "Towards a pragma-linguistic framework for the study of sensationalism in news headlines." *Discourse & communication* 7 (2): 173–197.
- Pantti, Mervi, Wahl-Jorgensen, Karin, and Cottle, Simon. 2012. *Disasters and the Media*. New York, United States of America: Peter Lang Verlag.
- Scheufele, DA. 1999. "Framing as a theory of media effects." *Journal of Communication* 49 (1): 103–122. <https://doi.org/10.1111/j.1460-2466.1999.tb02784.x>.
- Straka, Milan and Straková, Jana. 2017. "Tokenizing, POS Tagging, Lemmatizing and Parsing UD 2.0 with UDPipe." In *Proceedings of the CoNLL 2017 Shared Task: Multilingual Parsing from Raw Text to Universal Dependencies*, edited by Jan Hajič and Dan Zeman, 88–99. Association for Computational Linguistics. <https://doi.org/10.18653/v1/K17-3009>.
- Taulé, Mariona, Martí, M. Antònia, and Recasens, Marta. 2008. "AnCora: Multilevel Annotated Corpora for Catalan and Spanish." In *Proceedings of the Sixth International Conference on Language Resources and Evaluation (LREC'08)*, edited by Nicoletta Calzolari, Khalid Choukri, Bente Maegaard, Joseph Mariani, Jan Odijk, Stelios Piperidis, and Daniel Tapias. Marrakech, Morocco: European Language Resources Association (ELRA), May.
- Thompson, John Bodine. 1995. *The Media and Modernity: A Social Theory of the Media*. Stanford University Press.
- Van Dijk, Teun. 1988. *News as Discourse*. Hillsdale, NJ: Lawrence Erlbaum Associates.